

The Post Correspondence Problem

Some undecidable problems for context-free languages:

- Is $L(G_1) \cap L(G_2) = \emptyset$?

G_1, G_2 are context-free grammars

- Is context-free grammar G ambiguous?

We need a tool to prove that the previous problems for context-free languages are undecidable:

The Post Correspondence Problem

The Post Correspondence Problem

Input: Two sets of n strings

$$A = w_1, w_2, \dots, w_n$$

$$B = v_1, v_2, \dots, v_n$$

There is a Post Correspondence Solution
if there is a sequence $i, j, \dots, k$ such that:

PC-solution: $w_i w_j \cdots w_k = v_i v_j \cdots v_k$

Indices may be repeated or omitted

Example:

	w_1	w_2	w_3
$A:$	100	11	111

	v_1	v_2	v_3
$B:$	001	111	11

PC-solution: 2,1,3

$$w_2 w_1 w_3 = v_2 v_1 v_3$$

11100111

Example:

	w_1	w_2	w_3
$A:$	00	001	1000

	v_1	v_2	v_3
$B:$	0	11	011

There is no solution

Because total length of strings from B
is smaller than total length of strings from A

The Modified Post Correspondence Problem

Inputs: $A = w_1, w_2, \dots, w_n$

$$B = v_1, v_2, \dots, v_n$$

MPC-solution: $1, i, j, \dots, k$

$$w_1 w_i w_j \cdots w_k = v_1 v_i v_j \cdots v_k$$

Example:

	w_1	w_2	w_3
$A:$	11	111	100

	v_1	v_2	v_3
$B:$	111	11	001

MPC-solution: 1,3,2

$$w_1 w_3 w_2 = v_1 v_3 v_2$$

11100111

We will show:

1. The MPC problem is undecidable
(by reducing the membership to MPC)
2. The PC problem is undecidable
(by reducing MPC to PC)

Theorem: The MPC problem is undecidable

Proof: We will reduce the membership problem to the MPC problem

Membership problem

Input: Turing machine M
string w

Question: $w \in L(M)$?

Undecidable

Membership problem

Input: unrestricted grammar G
string w

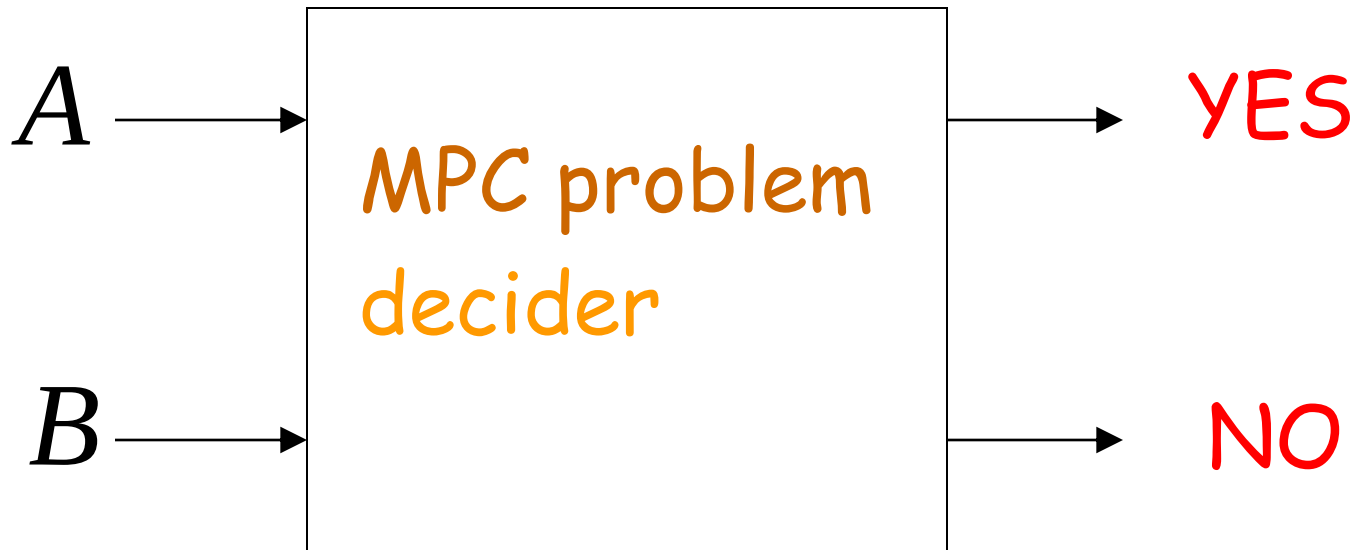
Question: $w \in L(G)$?

Undecidable

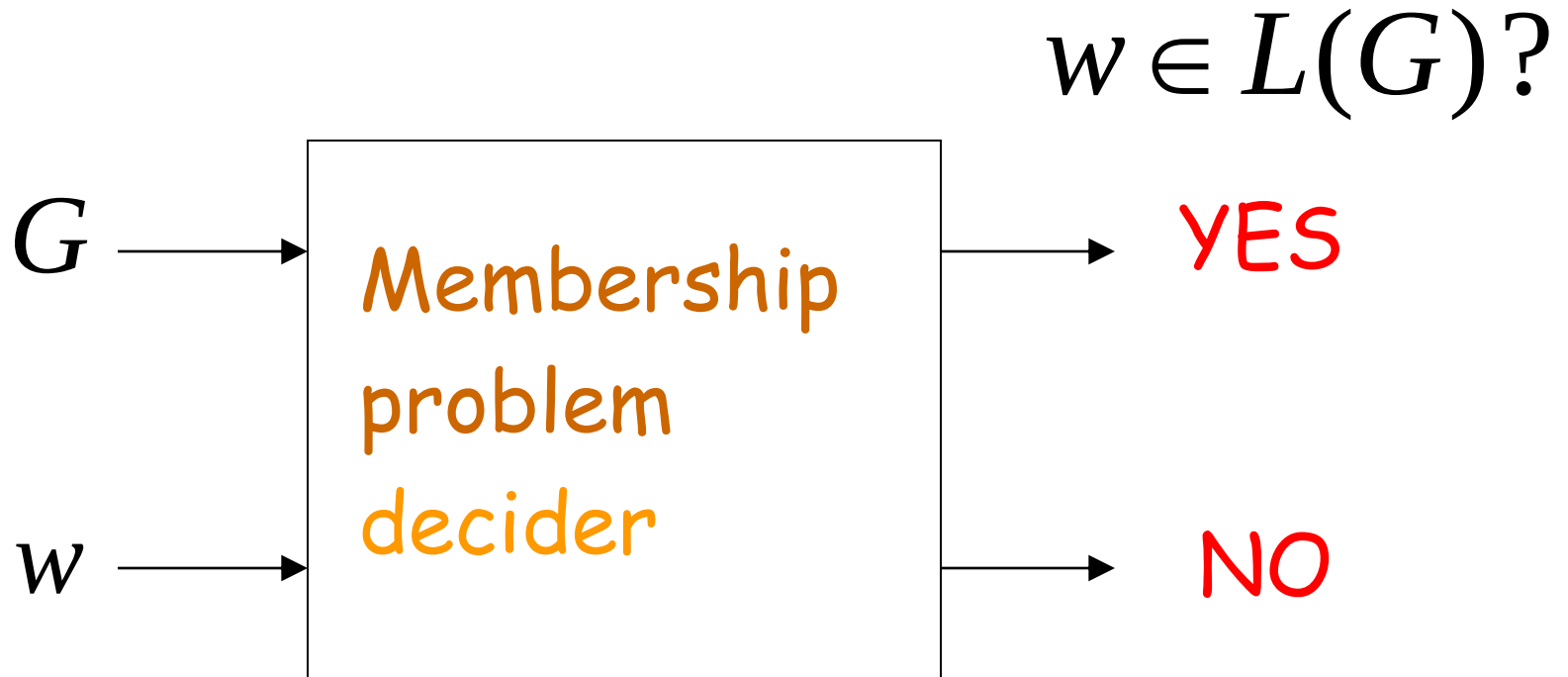
Suppose we have a decider for
the MPC problem

String Sequences

MPC solution?

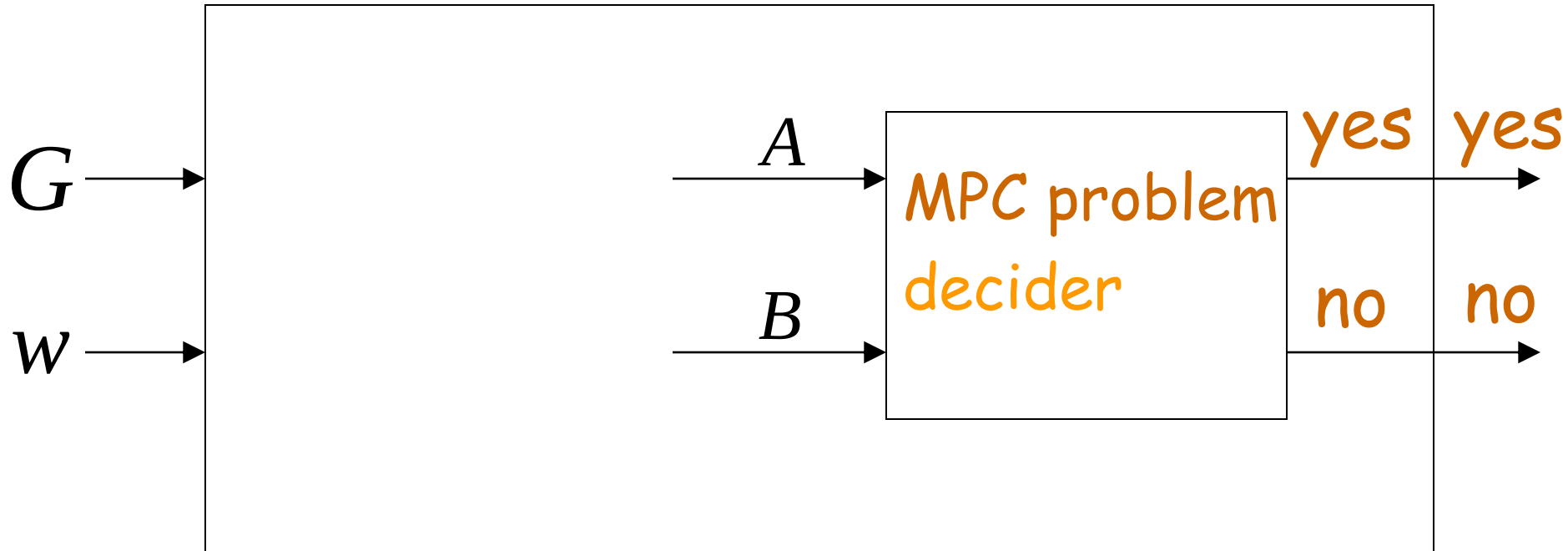


We will build a decider for
the membership problem



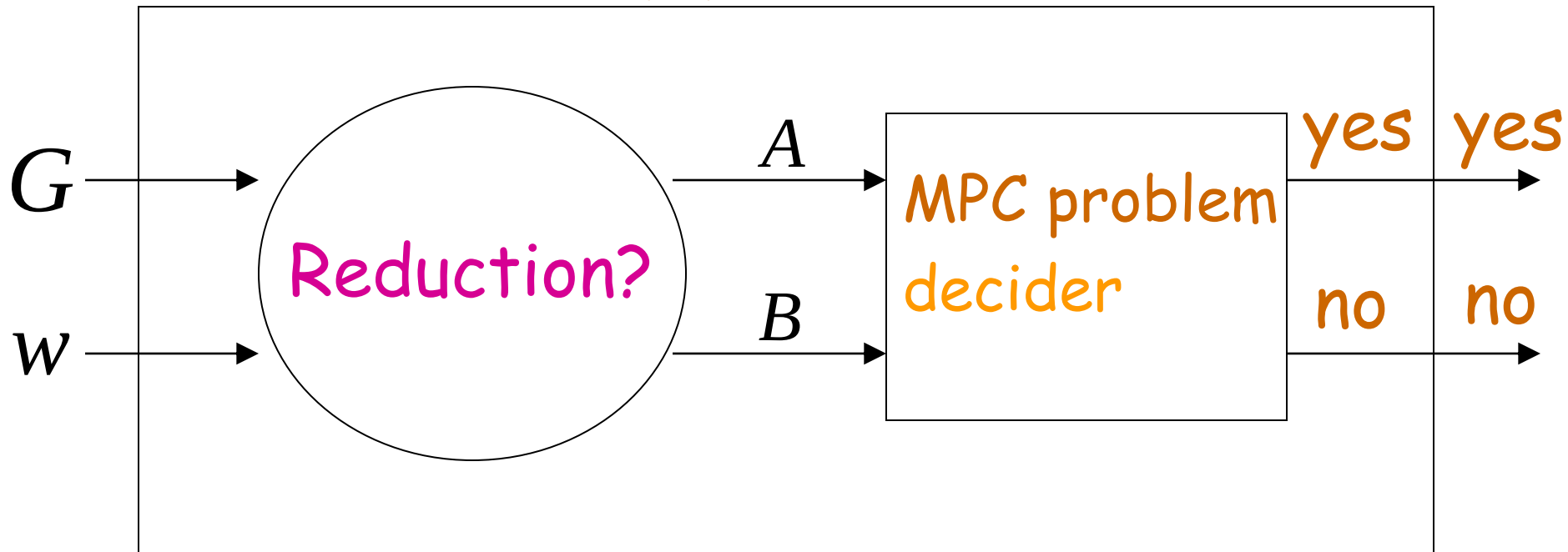
The reduction of the membership problem
to the MPC problem:

Membership problem decider



We need to convert the input instance of one problem to the other

Membership problem decider



Reduction:

Convert grammar G and string w
to sets of strings A and B

Such that:

G generates w  There is an MPC
solution for A, B

A	B	Grammar G
$FS \Rightarrow$	F	S : start variable F : special symbol
a	a	For every symbol a
V	V	For every variable V

A	B	Grammar G
E	$\Rightarrow wE$	string w E : special symbol
y	x	For every production $x \rightarrow y$
$\Rightarrow$	$\Rightarrow$	

Example:

Grammar G :

$$S \rightarrow aABb \mid Bbb$$
$$Bb \rightarrow C$$
$$AC \rightarrow aac$$

String $w = aaac$

A B $w_1 : FS \Rightarrow$ $v_1 : F$ $w_2 : a$ $v_2 : a$ $w_3 : b$ $v_3 : b$ c c $\vdots A$ $\vdots A$ B B C C $w_8 : S$ $v_8 : S$

A B $w_9 :$ E $v_9 :$ $\Rightarrow aaacE$ $aABb$ S Bbb S $\vdots$ $\vdots$ C Bb aac AC $w_{14} :$ $\Rightarrow$ $v_{14} :$ $\Rightarrow$

Grammar G : $S \rightarrow aABb \mid Bbb$

$Bb \rightarrow C$

$AC \rightarrow aac$

$aaac \in L(G)$: $S \Rightarrow aABb \Rightarrow aAC \Rightarrow aaac$

Derivation: S

$$S \rightarrow aABb \mid Bbb$$

$$Bb \rightarrow C$$

$$AC \rightarrow aac$$

$$A : \quad w_1$$
$$\underbrace{\quad \quad \quad}_{F \quad S \Rightarrow}$$

$$B : \quad v_1$$

Derivation:

$$S \Rightarrow aABb$$

$$S \rightarrow aABb \mid Bbb$$

$$Bb \rightarrow C$$

$$AC \rightarrow aac$$

$$A : \quad \begin{array}{c} w_1 \qquad w_{10} \\ \underbrace{\qquad\qquad\qquad} \quad \underbrace{\qquad\qquad\qquad} \\ F \quad S \Rightarrow a \quad A \quad B \quad b \\ \underbrace{\quad} \quad \underbrace{\quad} \\ \downarrow \quad \downarrow \end{array}$$

$$B : \quad v_1 \quad v_{10}$$

Derivation:

$$S \Rightarrow aABb \Rightarrow aAC$$

$$S \rightarrow aABb \mid Bbb$$

$$Bb \rightarrow C$$

$$AC \rightarrow aac$$

$$A : \quad w_1 \quad w_{10} \quad w_{14} \quad w_2 \quad w_5 \quad w_{12}$$

$$\begin{array}{ccccccc} \underbrace{\quad \quad \quad} & \underbrace{\quad \quad \quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} \\ F & S & \Rightarrow a & A & B & b \Rightarrow a & A & C \\ \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} & \underbrace{\quad} \end{array}$$

$$B : \quad v_1 \quad v_{10} \quad v_{14} \quad v_2 \quad v_5 \quad v_{12}$$

Derivation:

$$S \Rightarrow aABb \Rightarrow aAC \Rightarrow aaac$$

$$S \rightarrow aABb \mid Bbb$$

$$Bb \rightarrow C$$

$$AC \rightarrow aac$$

A : $w_1 \quad w_{10} \quad w_{14} \quad w_2 \quad w_5 \quad w_{12} \quad w_{14} \quad w_2 \quad w_{13}$

$F \quad S \Rightarrow a \quad A \quad B \quad b \Rightarrow a \quad A \quad C \Rightarrow a \quad a \quad a \quad c \quad E$

B : $v_1 \quad v_{10} \quad v_{14} \quad v_2 \quad v_5 \quad v_{12} \quad v_{14} \quad v_2 \quad v_{13}$

Derivation:

$$S \Rightarrow aABb \Rightarrow aAC \Rightarrow aaac$$

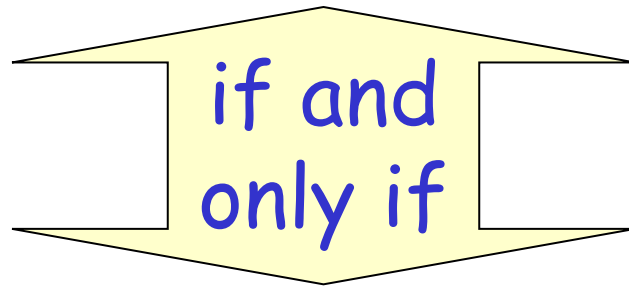
$$S \rightarrow aABb \mid Bbb$$

$$Bb \rightarrow C$$

$$AC \rightarrow aac$$

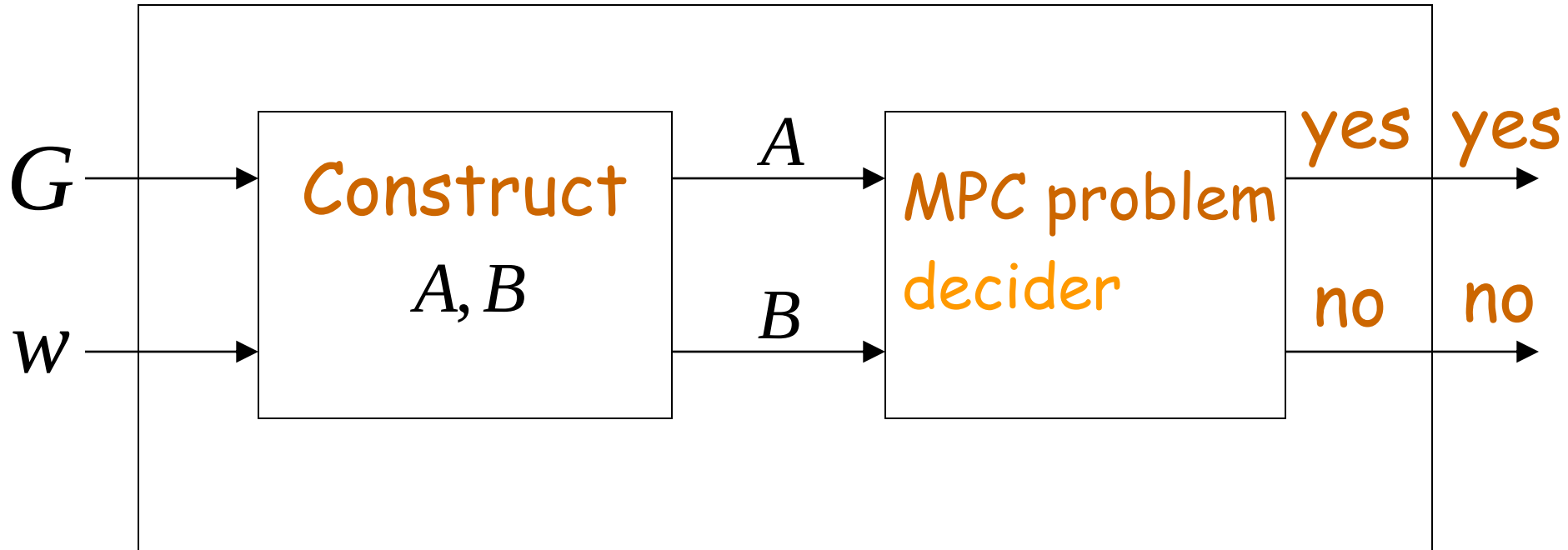
$$\begin{array}{c}
 \mathbf{A} : \quad w_1 \quad \quad w_{10} \quad w_{14} \quad w_2 \quad w_5 \quad w_{12} w_{14} \quad w_2 \quad w_{13} \quad w_9 \\
 \begin{array}{c}
 \underbrace{\quad \quad \quad} \underbrace{\quad \quad \quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad \quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad \quad} \underbrace{\quad \quad \quad} \underbrace{\quad} \\
 F \quad S \Rightarrow a \quad A \quad B \quad b \Rightarrow a \quad A \quad C \quad \Rightarrow a \quad a \quad a \quad c \quad E \\
 \underbrace{\quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad \quad} \underbrace{\quad} \underbrace{\quad} \underbrace{\quad \quad} \underbrace{\quad \quad \quad \quad \quad \quad \quad} \\
 \end{array} \\
 \mathbf{B} : v_1 \quad v_{10} \quad v_{14} \quad v_2 \quad v_5 \quad v_{12} \quad v_{14} \quad v_2 \quad v_{13} \quad \quad \quad v_9
 \end{array}$$

(A, B) has an MPC-solution



$w \in L(G)$

Membership problem decider



Since the membership problem is undecidable,
The MPC problem is undecidable

END OF PROOF

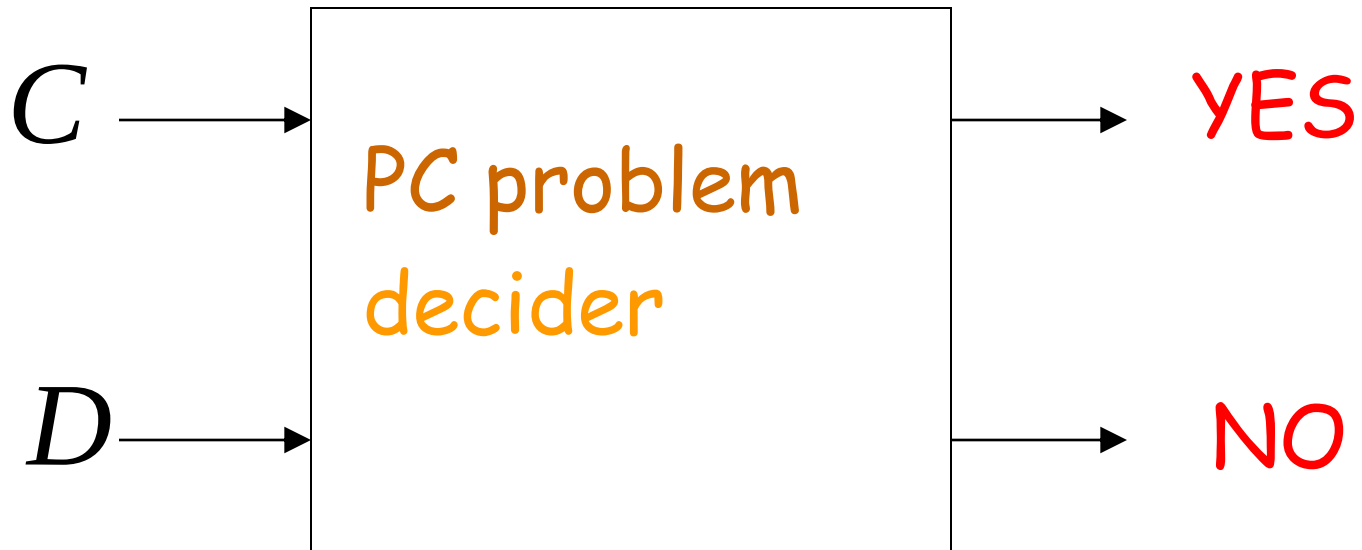
Theorem: The PC problem is undecidable

Proof: We will reduce the MPC problem
to the PC problem

Suppose we have a decider for
the PC problem

String Sequences

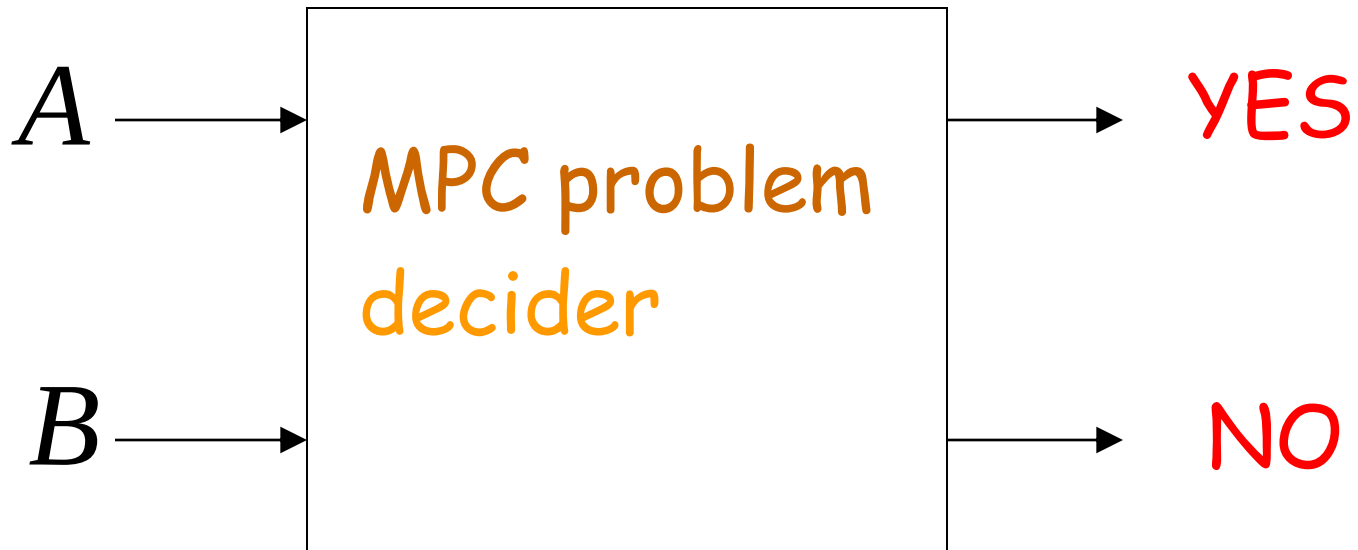
PC solution?



We will build a decider for
the MPC problem

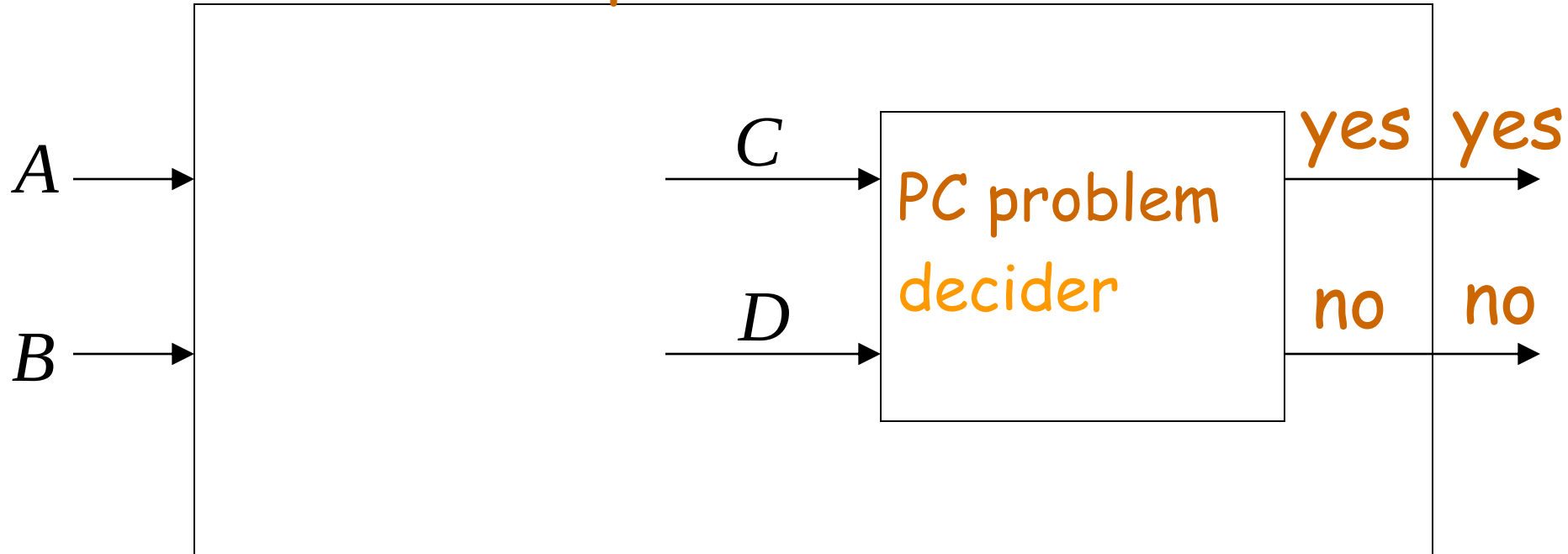
String Sequences

MPC solution?



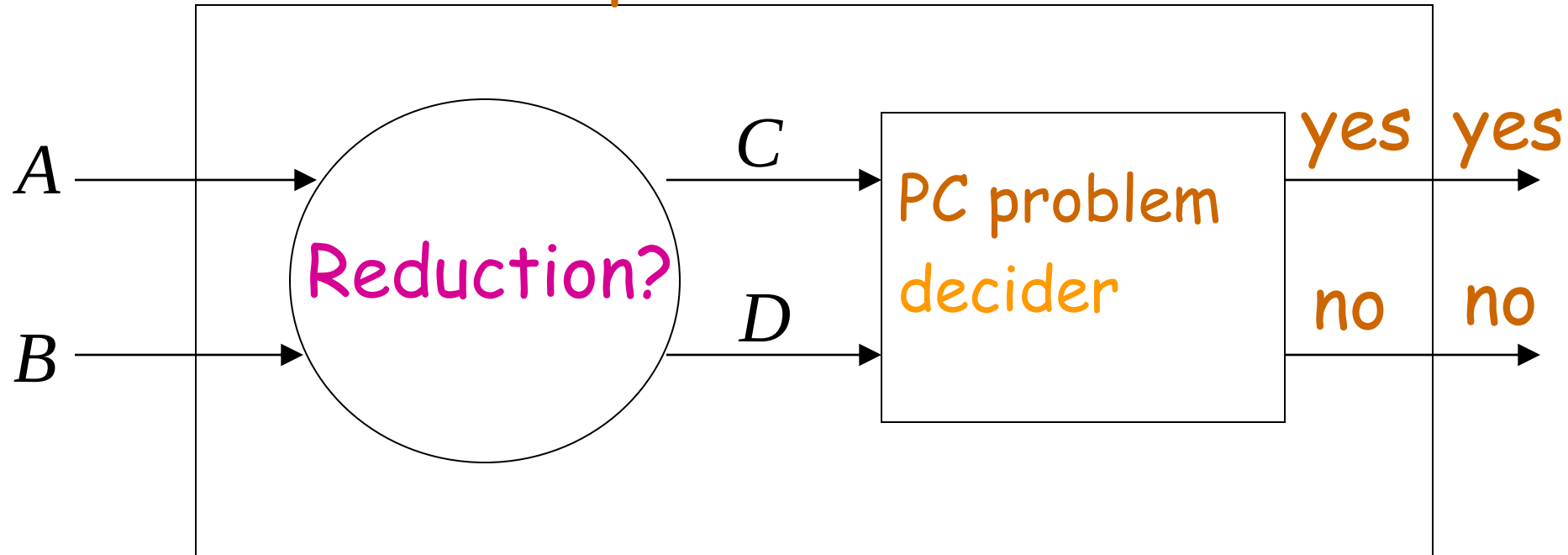
The reduction of the MPC problem to the PC problem:

MPC problem decider



We need to convert the input instance of one problem to the other

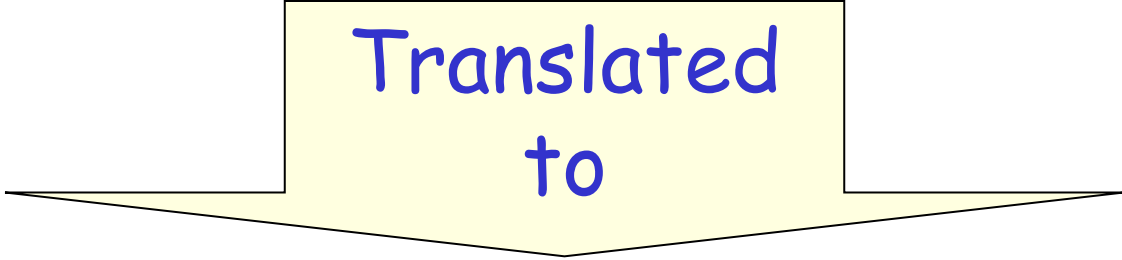
MPC problem decider



A, B : input to the MPC problem

$$A = w_1, w_2, \dots, w_n$$

$$B = v_1, v_2, \dots, v_n$$



Translated
to

C, D : input to the PC problem

$$C = w'_1, \dots, w'_n, w'_{n+1}$$

$$D = v'_1, \dots, v'_n, v'_{n+1}$$

A

$$w_i = \sigma_1 \sigma_2 \cdots \sigma_k$$

For each i

 C

$$w'_i = \sigma_1 * \sigma_2 * \cdots \sigma_k *$$

replace $w'_1 = * w'_1$

$$w'_{n+1} = \diamond$$

 B

$$v_i = \pi_1 \pi_2 \cdots \pi_k$$

For each i

 D

$$v'_i = * \pi_1 * \pi_2 * \cdots * \pi_k$$

$$v'_{n+1} = * \diamond$$

PC-solution

$$\overset{C}{w_1' w_i' \cdots w_k' w_{n+1}'} = \overset{D}{v_1' v_i' \cdots w_k' v_{n+1}'}$$

Has to start with
These strings

C PC-solution

D

$$w_1' w_i' \cdots w_k' w_{n+1}' = v_1' v_i' \cdots w_k' v_{n+1}'$$

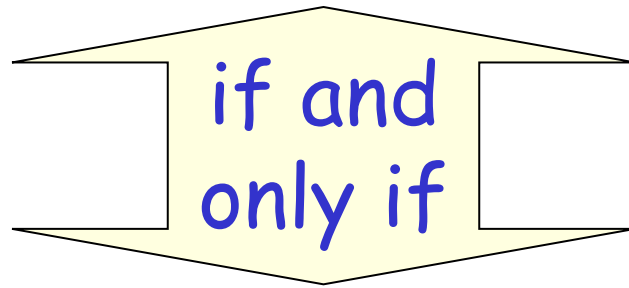
A

B

$$w_1 w_i \cdots w_k = v_1 v_i \cdots v_k$$

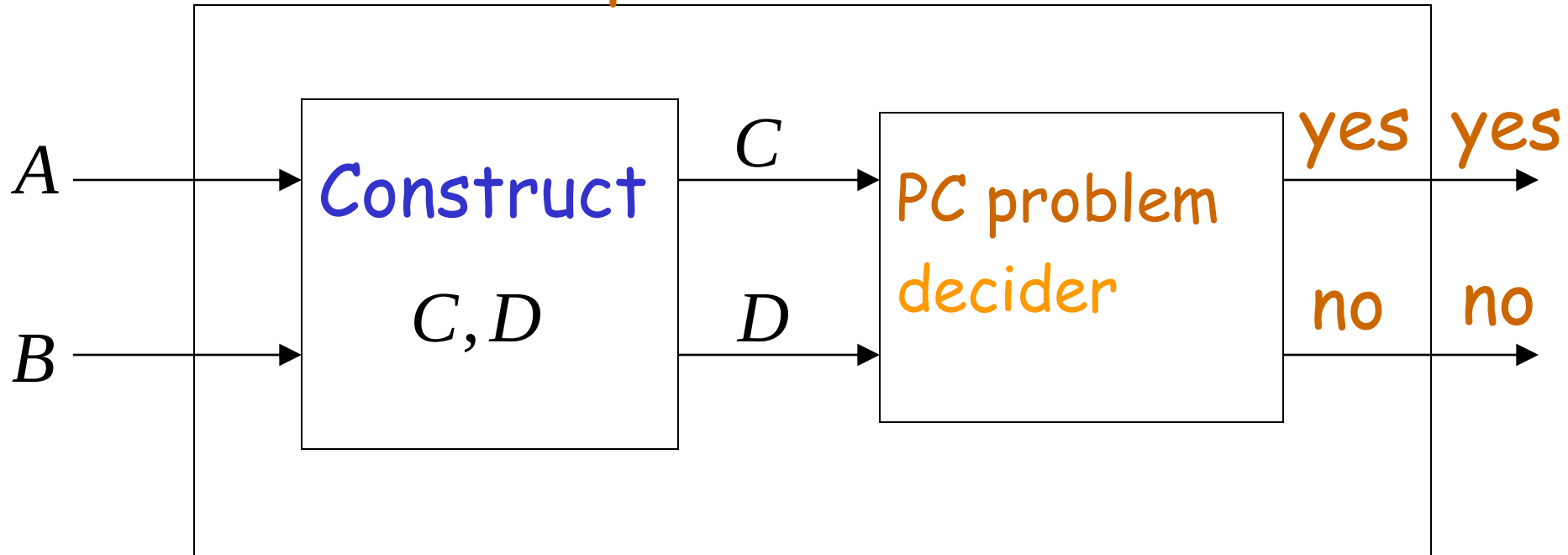
MPC-solution

C, D has a PC solution



A, B has an MPC solution

MPC problem decider



Since the MPC problem is undecidable,
The PC problem is undecidable

END OF PROOF

Some undecidable problems for context-free languages:

- Is $L(G_1) \cap L(G_2) = \emptyset$?
 G_1, G_2 are context-free grammars
- Is context-free grammar G ambiguous?

We reduce the PC problem to these problems

Theorem: Let G_1, G_2 be context-free grammars. It is undecidable to determine if

$$L(G_1) \cap L(G_2) = \emptyset$$

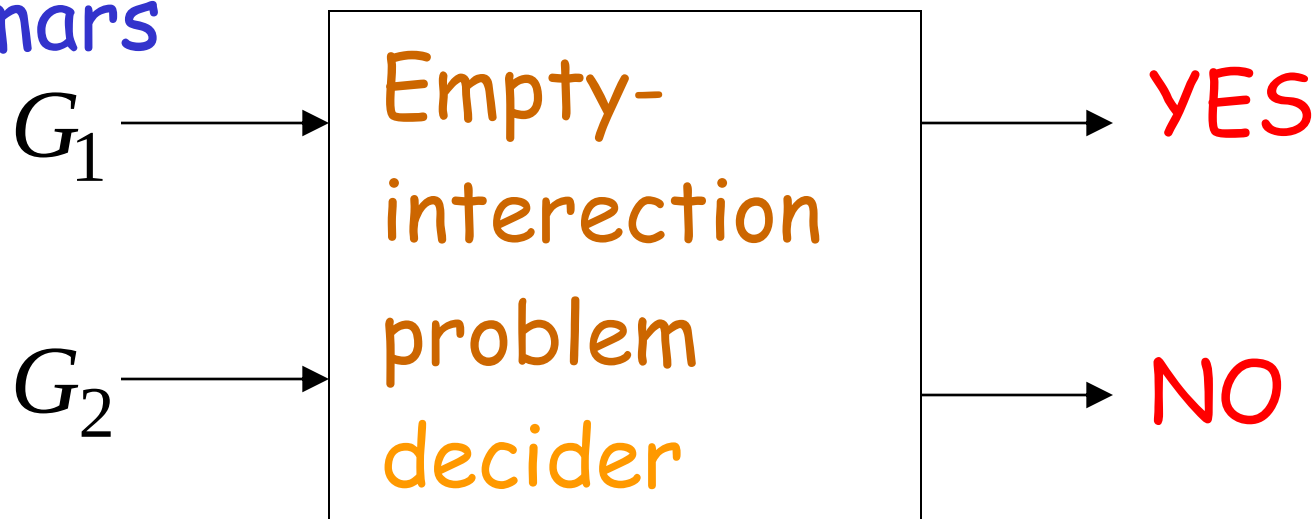
(intersection problem)

Proof: Reduce the PC problem to this problem

Suppose we have a decider for the intersection problem

Context-free
grammars

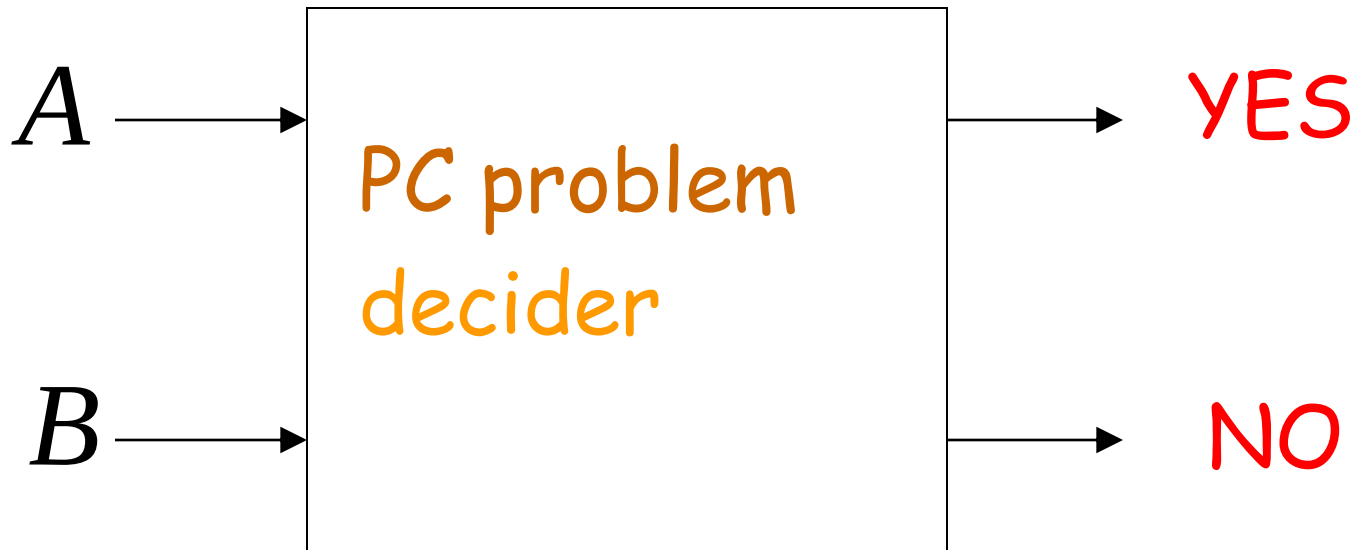
$$L(G_1) \cap L(G_2) = \emptyset?$$



We will build a decider for
the PC problem

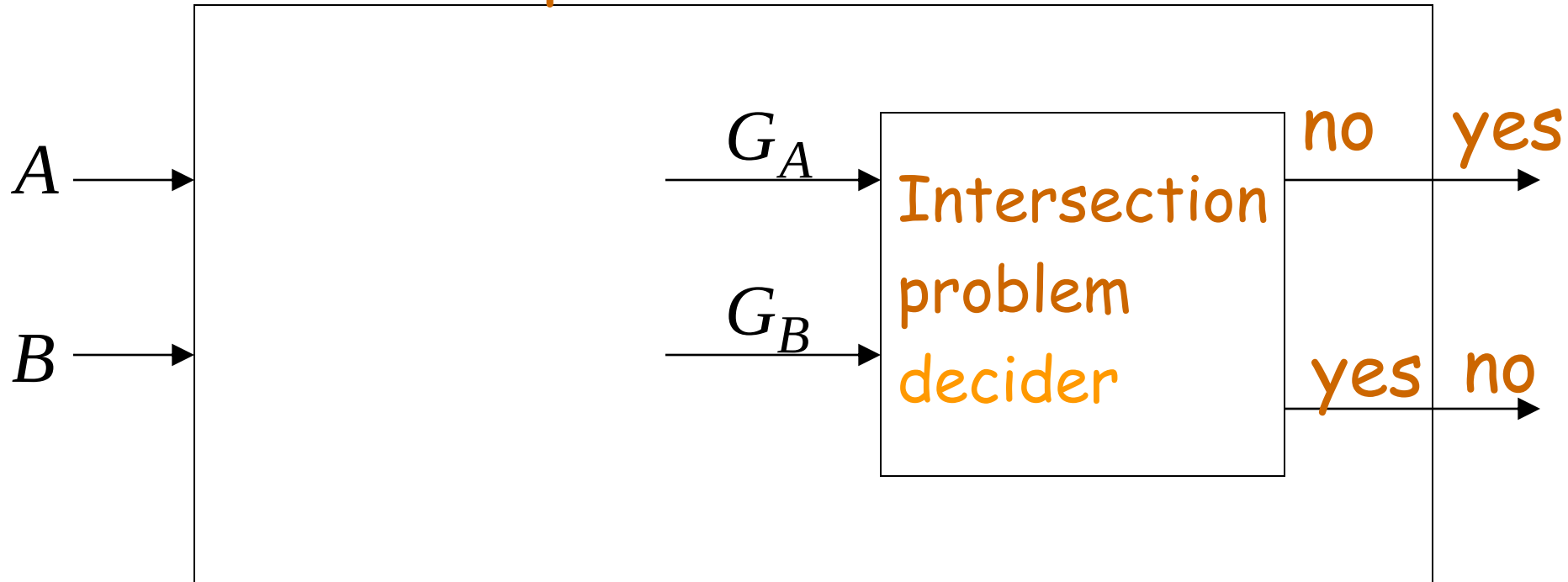
String Sequences

PC solution?



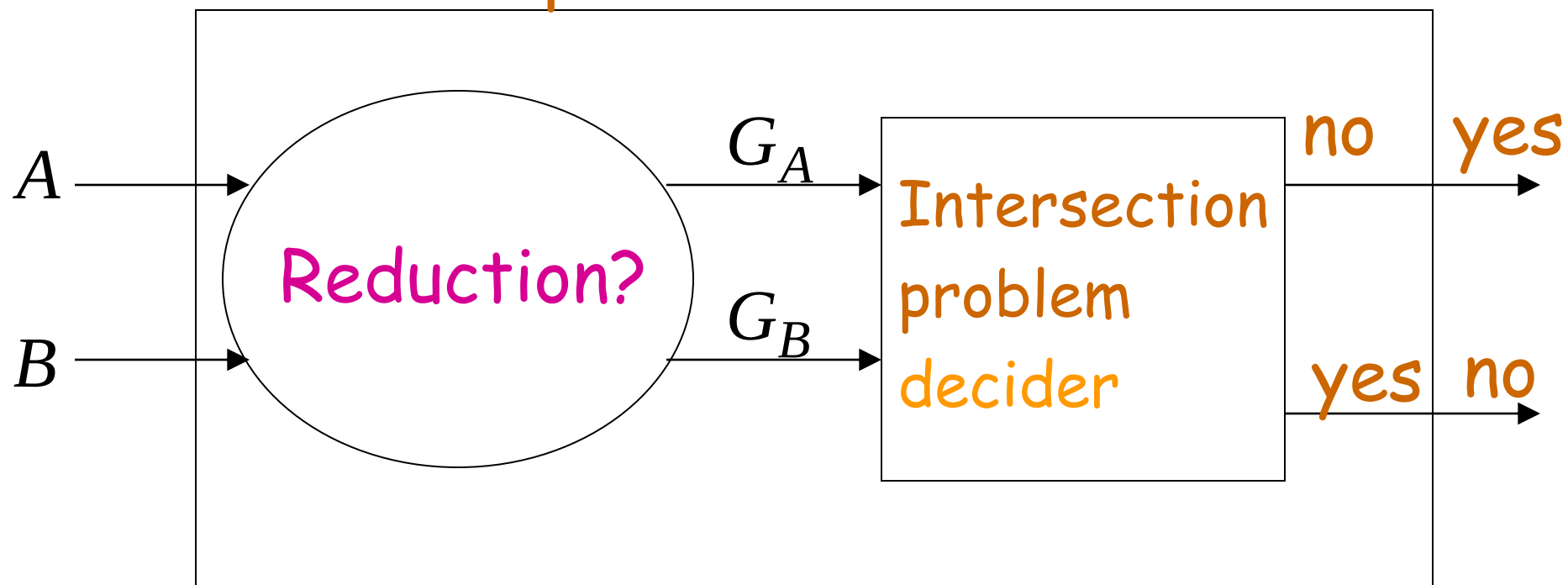
The reduction of the PC problem
to the empty-intersection problem:

PC problem decider



We need to convert the input instance of one problem to the other

PC problem decider



Introduce new unique symbols: $a_1, a_2, \dots, a_n$

$$A = w_1, w_2, \dots, w_n$$

$$L_A = \{s : s = w_i w_j \cdots w_k a_k \cdots a_j a_i\}$$

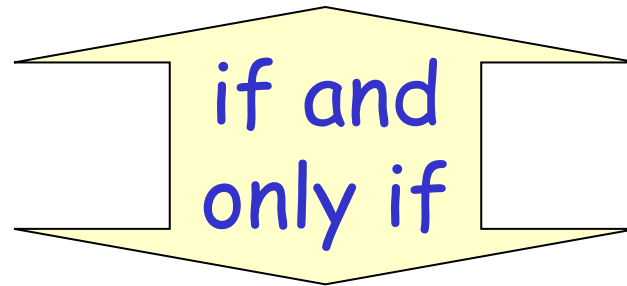
Context-free grammar $G_A: S_A \rightarrow w_i S_A a_i \mid w_i a_i$

$$B = v_1, v_2, \dots, v_n$$

$$L_B = \{s : s = v_i v_j \cdots v_k a_k \cdots a_j a_i\}$$

Context-free grammar $G_B: S_B \rightarrow v_i S_B a_i \mid v_i a_i$

(A, B) has a PC solution



$$L(G_A) \cap L(G_B) \neq \emptyset$$

$$L(G_1) \cap L(G_2) \neq \emptyset$$

$$S = w_i w_j \cdots w_k a_k \cdots a_j a_i$$

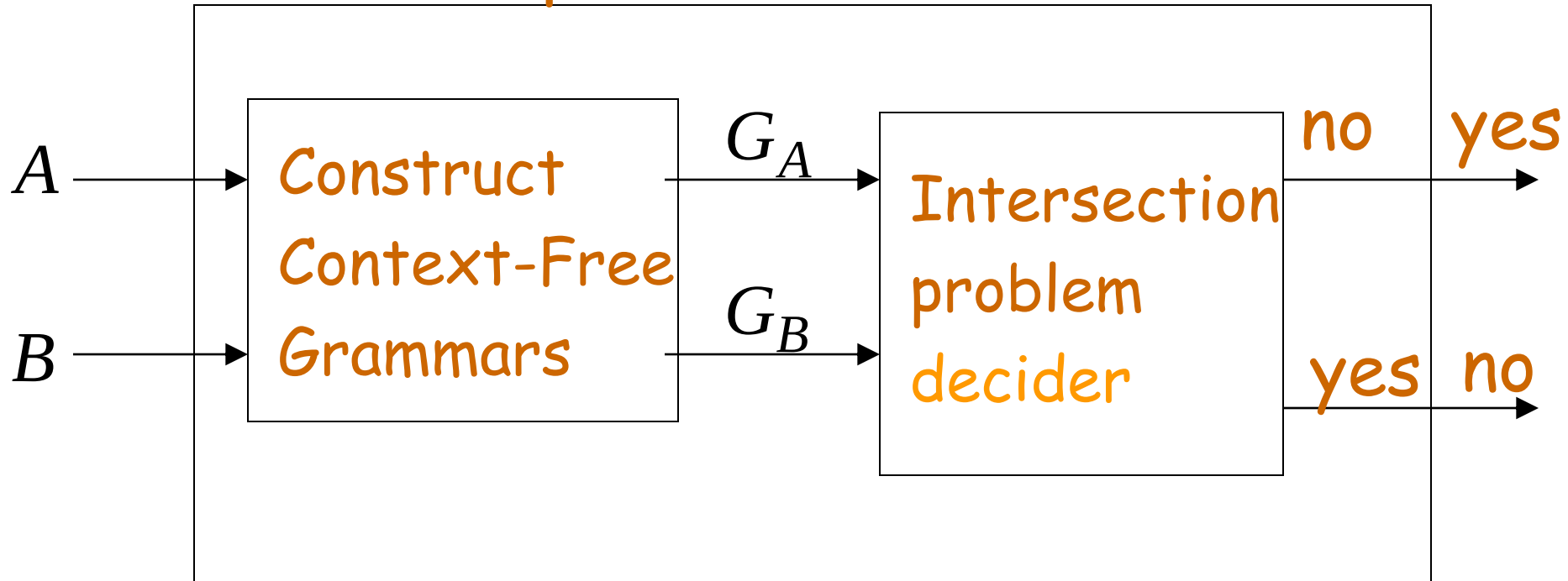
$$S = v_i v_j \cdots v_k a_k \cdots a_j a_i$$

Because $a_1, a_2, \dots, a_n$ are unique

There is a PC solution:

$$w_i w_j \cdots w_k = v_i v_j \cdots v_k$$

PC problem decider



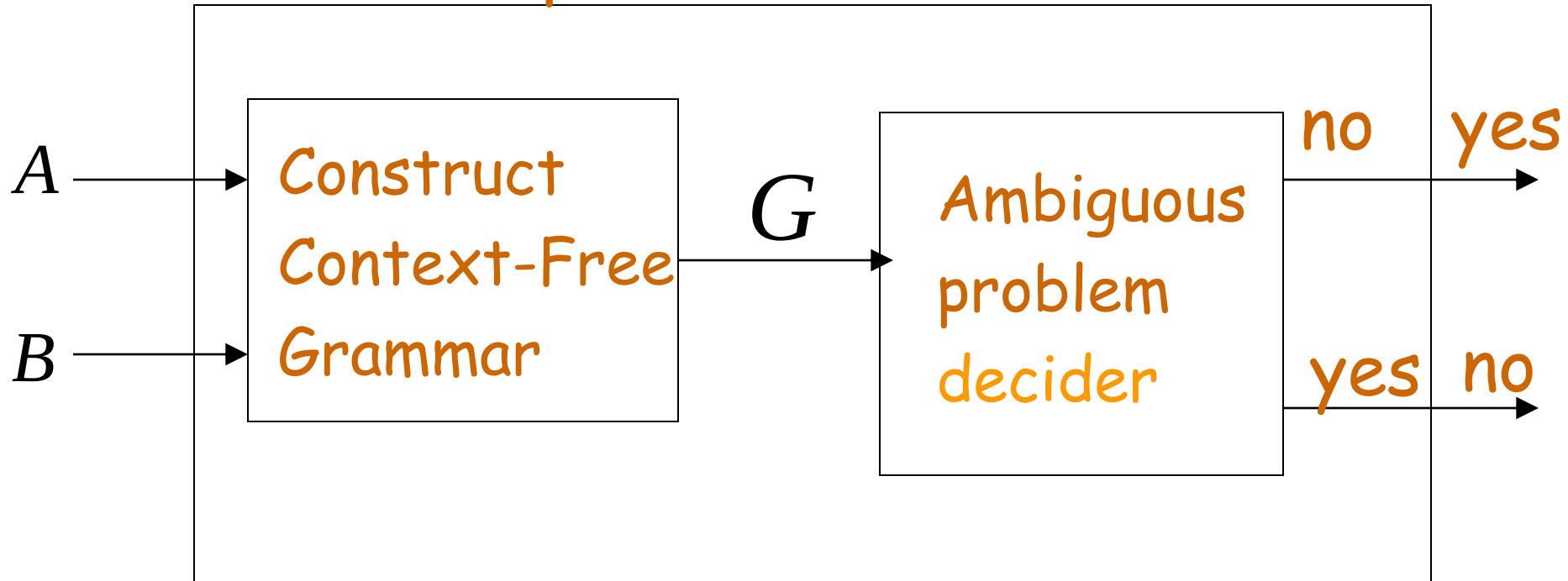
Since PC is undecidable,
the Intersection problem is undecidable

END OF PROOF

Theorem: For a context-free grammar G ,
it is undecidable to determine
if G is ambiguous

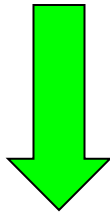
Proof: Reduce the PC problem
to this problem

PC problem decider



S_A start variable of G_A

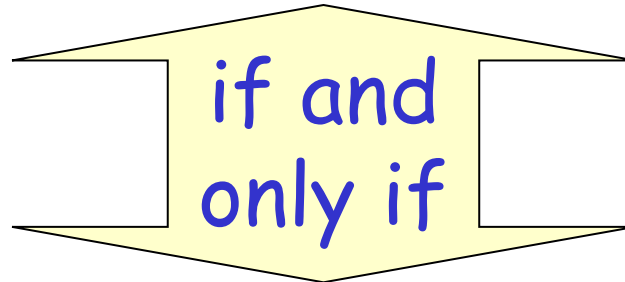
S_B start variable of G_B



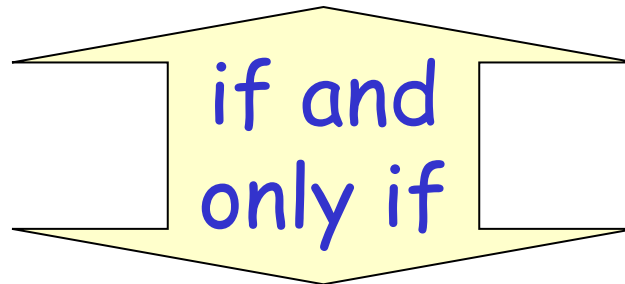
S start variable of G

$$S \rightarrow S_A \mid S_B$$

(A, B) has a PC solution



$$L(G_A) \cap L(G_B) \neq \emptyset$$



G is ambiguous